

Report for Computer Vision in Archaeology training school of ATRIUM project (4-18 September 2026)

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This report provides an overview of the ATRIUM Computer Vision in Archaeology Training School, that took place in Brno (Czech Republic) between 14th to 19th September 2026. The program covered a variety of topics through a combination of theoretical lectures and practical sessions. Experts in the field introduced participants to a range of tools, methods, and workflows for applying computer vision and machine learning to archaeological research.



Day 1. Kick-off, Introduction to Computer Vision, Agents and Python

The first day began with a welcome session for the participants, followed by organizational information, an introduction to the course programme, and self-introductions. The day continued with lectures on the applications of computer vision, the use and parameter settings of coding agents, and an introduction to basic programming in Python. The day concluded with a social icebreaker event on the terrace of the Institute of Archaeology of the Czech Academy of Sciences, providing an opportunity for participants to get to know one another in a relaxed environment.



Figure 1. Ice breaker event in the terrace of the institute.

Day 2. Artifact images and Participants' Presentation

The second day focused on the first steps in applying computer vision to archaeological images. The theoretical session introduced participants to the image-processing pipeline, including image annotation and dataset preparation for training predictive algorithms using software such as CVAT.ai. A practical session followed with the participants annotating

images of archaeological artifacts provided by the lecturer, meanwhile participants with their own datasets had the opportunity to annotate their images. This was followed by a practical session in which participants annotated images of archaeological artifacts provided by the lecturer. Participants who had their own datasets were also given the opportunity to work with and annotate their own images. Personally, I was able to annotate microscope images of bone tissue by creating two categories, “Secondary osteons” and “Osteon Fragments.” I then downloaded the resulting XML annotation file, which can be used for training a computer vision algorithm. After the main session, in the evening we attended participants’ presentations getting acquaintance with their field of study and their projects related to computer vision. In the evening, we attended the participants’ presentations, which provided an opportunity to learn more about their respective fields of study and their research projects involving computer vision. Following the presentations, we took part in a guided tour of Brno, where we learned about local myths and visited some of the city's most important sights. The day concluded with a gathering where we had the opportunity to taste local beers.



Figure 2. Cathedral of Brno and beer from the evening gathering

Day 3. Microscopic Images

The third day focused on the processing and analysis of microscopic images. Using a dataset of stone-tool microwear, we learned how to pre-process images through standardization, normalization, and noise reduction before applying machine-learning methods. The first part of the day consisted of a comprehensive theoretical session covering the different stages of the workflow and explaining the purpose and importance of each step. The lecture was interactive, with questions and exercises encouraging participants to actively engage with the material. The second part of the day was dedicated to practical work, during which we trained and tested a machine-learning model. During the lunch break, we had the opportunity to visit the institute's laboratories and learn more about its facilities, research activities, and ongoing projects. The day concluded with a social event at the institute, where we were able to taste a variety of local foods and snacks.



Figure 3. Welcome event

Day 4. Satellite images and Rock art

The fourth day was divided into two main topics: satellite imagery and rock art. The first session introduced us to the processing of satellite

images for the extraction of geospatial data that as subsequently used to train machine-learning algorithms. The second session showed us how to identify patterns of rock art and to enhance them using CLAHE for training an identification model. The workflow covered the preparation and organization of training, validation, and test datasets, followed by model training and the subsequent detection of cupmarks in previously unseen images. The day concluded with a visit to the institute's archives, providing further insight into the resources and research infrastructure available at the institution.

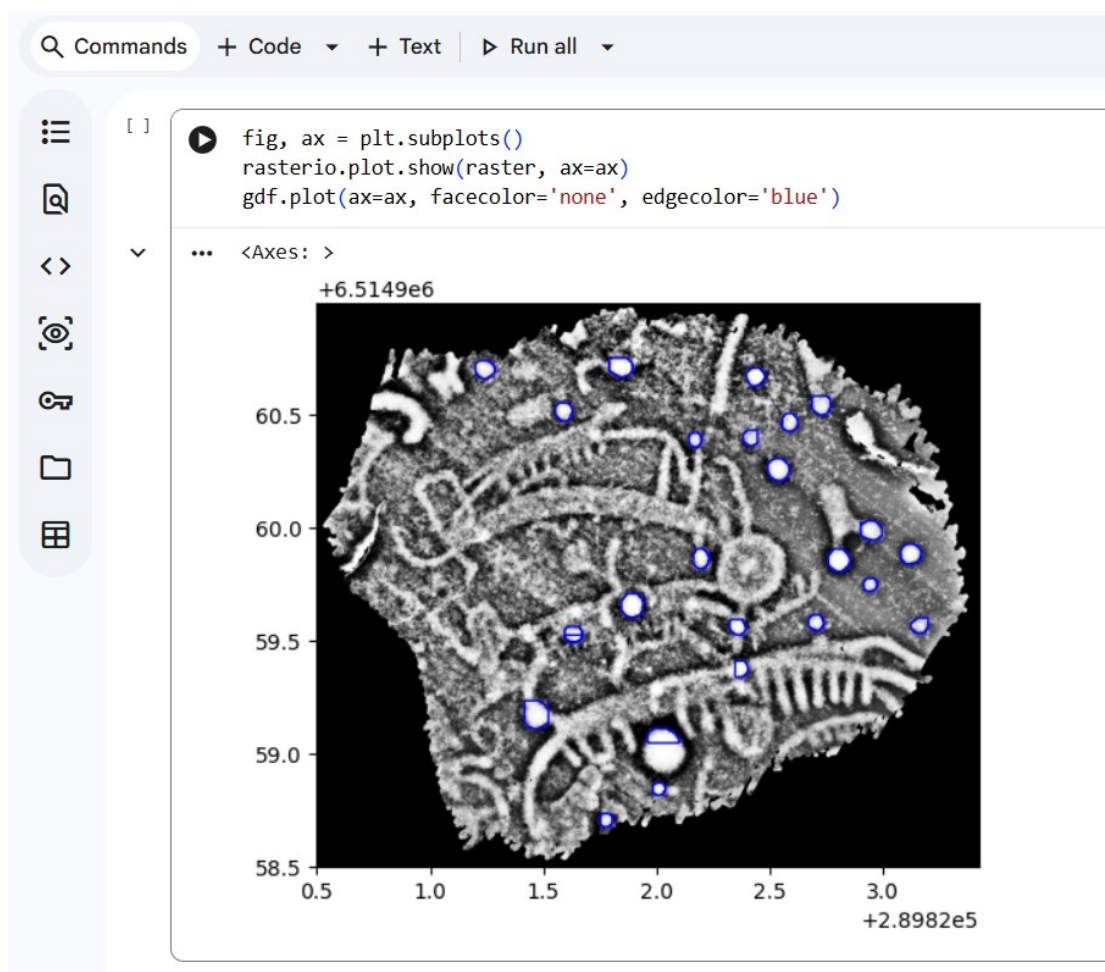


Figure 4. Rock art image with detected cupmarks.

Day 5. Coins dataset and Closure

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into the resources and research infrastructure available at the institution. The program concluded with a final discussion, during which participants reflected on the approaches and techniques presented throughout the training and considered their relevance and potential applications in future archaeological research.



Figure 5. Group photo of participants, lecturers and organizing members.

Closing remarks

Overall, the ATRIUM training school expanded my understanding of the possibilities and challenges associated with applying computer vision to archaeological material. The skills and knowledge gained during the program, particularly through the practical work with image annotation and model training, can be valuable for future research and may contribute to the development of more efficient and reproducible approaches to archaeological analysis. Last but not least I had the opportunity to meet new people beside my field of study and expand my network.



Figure 6. Photo of me with colleagues Zabya and Michelle traveling after the closure of the ATRIUM training school.